

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

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Section	SECTION 01
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Case Study	WBL PROJECT
System Name	Koperasri KADA Online system
Group Name	Technicrab

SECTION C (SQL AND INTERFACE IMPLEMENTATION)		
3.1	SQL Statement (DDL & DML)	
		<< Provide the SQL statements (DDL and DML) for a set of queries (minimum 10 queries) that can be implemented in your proposed module. In addition, please include the screenshots of the output/results for each SQL statement.>>
3.2	Interface Implementation	
		<< Provide at least THREE (3) interface design reflecting queries in Subsection 3.1.>>
SECTION D (CONCLUSION AND REFLECTION)		
4.1	Conclusion	
		<< Provide achievement(s) that have been achieved in your module and suggestion(s) to further improvise in future .>>
4.2	Reflection	
		<< Provide your reflection based on your WBL project.>>

SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 SQL Statement (DDL & DML)

a) DDL

```
CREATE TABLE `tb_deduction` (  
  `Deduct_ID` int (10) NOT NULL,  
  `employeeID` int (4) NOT NULL,  
  `DeducType_ID` int (10) NOT NULL,  
  `Deduct_Amt` int (10) NOT NULL,  
  `Deduct_date` date NOT NULL DEFAULT current_timestamp ()  
)
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	Deduct_ID 	int(10)			No	None		AUTO_INCREMENT	 Change  Drop  More
2	employeeID 	int(4)			No	None			 Change  Drop  More
3	DeducType_ID 	int(10)			No	None			 Change  Drop  More
4	Deduct_Amt	int(10)			No	None			 Change  Drop  More
5	Deduct_date	date			No	None			 Change  Drop  More

```
ALTER TABLE `tb_deduction`  
  ADD PRIMARY KEY (`Deduct_ID`),  
  ADD KEY `deduction_member` (`employeeID`),  
  ADD KEY `deduction_type` (`DeducType_ID`);
```

Your SQL query has been executed successfully.

```
ALTER TABLE `tb_deduction` ADD CONSTRAINT `deduction_member` FOREIGN KEY (`employeeID`) REFERENCES `tb_member` (`employeeID`) ON DELETE RESTRICT ON UPDATE RESTRICT; ALTER TABLE `tb_deduction` ADD CONSTRAINT `deduction_type` FOREIGN KEY (`DeducType_ID`) REFERENCES `tb_deduction_type` (`DeducType_ID`) ON DELETE RESTRICT ON UPDATE RESTRICT;
```

[\[Edit inline\]](#) [\[Edit\]](#) [\[Create PHP code\]](#)

b) DML

```
//fetch the member detail and fee detail
```

```
SELECT DISTINCT  
  m.employeeID,  
  m.memberName,  
  mf.modalShare,  
  mf.feeCapital,  
  mf.fixedDeposit,  
  mf.contribution,  
  mf.deposit  
FROM tb_member m  
LEFT JOIN tb_memberregistration_feesandcontribution mf  
  ON m.employeeID = mf.employeeID
```

//fetch the loan repayment

```
SELECT
    l.employeeID,
    l.loanType,
    l.monthlyInstallments,
    l.balance
FROM tb_loan l
JOIN tb_loanapplication la ON l.loanApplicationID = la.loanApplicationID
WHERE la.loanStatus = 'Diluluskan'
AND l.balance > 0
```

//Insert deduction record for loan repayment

```
INSERT INTO tb_deduction (employeeID, DeducType_ID, Deduct_Amt, Deduct_date)
VALUES (?, 6, ?, ?)
```

//Insert deduction record for other payment types

```
INSERT INTO tb_deduction (employeeID, DeducType_ID, Deduct_Amt, Deduct_date)
VALUES (?, ?, ?, ?)
```

//fetch the deduction type

```
SELECT * FROM tb_deduction_Type
```

//fetch loan detail

```
SELECT
    l.employeeID,
    m.memberName,
    l.loanType,
    l.monthlyInstallments,
    l.balance,
    la.loanStatus
FROM tb_loan l
JOIN tb_loanapplication la
    ON l.loanApplicationID = la.loanApplicationID
```

```

JOIN tb_member m
    ON l.employeeID = m.employeeID
WHERE la.loanStatus = 'Diluluskan';

//select deduction type
SELECT DeducType_ID, typeName
FROM tb_deduction_type;

//get total deduction
SELECT
    d.employeeID,
    dt.typeName,
    SUM(d.Deduct_Amt) as total_deduction
FROM tb_deduction d
JOIN tb_deduction_type dt
    ON d.DeducType_ID = dt.DeducType_ID
WHERE d.employeeID = ?
GROUP BY d.employeeID, dt.typeName;

```

3.2 Interface Implementation

Upload Payment Record

☐

Upload Selected

Transaction Date:

01/25/2025

☐ 322 Cheryl Cheong Kah Voon

Upload

Modal Share (Min: RM 1,000.00)	RM 1,000.00	2000
Fee Capital (Min: RM 0.00)	RM 0.00	0
Fixed Deposit (Min: RM 1,200.00)	RM 1,200.00	1200
Contribution (Min: RM 600.00)	RM 600.00	600

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LIM MEI MEI

Upload

Modal Share (Min: RM 300.00)	RM 300.00	300
Fee Capital (Min: RM 40.00)	RM 40.00	40
Fixed Deposit (Min: RM 40.00)	RM 40.00	40
Contribution (Min: RM 40.00)	RM 40.00	40
Deposit (Min: RM 40.00)	RM 40.00	40
Loan Repayment (AL-INAH) (Min: RM 750.00)	RM 750.00	750.00
Total Amount		RM 1,210.00

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Chau Ying Jia

Upload

Modal Share (Min: RM 300.00)	RM 300.00	300
Fee Capital (Min: RM 300.00)	RM 300.00	300
Fixed Deposit (Min: RM 300.00)	RM 300.00	300

The screenshot displays two web forms for financial management. The top form, titled 'CONTRIBUTION (MIN: RM 300.00)', contains three input fields: 'Deposit (Min: RM 300.00)' with a value of 300, 'Loan Repayment (AL-BAI) (Min: RM 100.10)' with a value of 100.10, and a 'Total Amount' of RM 1,600.10. The bottom form, titled '9976 GUI KAH SIN' with an 'Upload' button, contains six input fields: 'Modal Share (Min: RM 300.00)' with a value of 300, 'Fee Capital (Min: RM 20.00)' with a value of 20, 'Fixed Deposit (Min: RM 20.00)' with a value of 20, 'Contribution (Min: RM 20.00)' with a value of 20, 'Deposit (Min: RM 20.00)' with a value of 20, and 'Loan Repayment (AL-BAI) (Min: RM 250.00)' with a value of 250.00. The 'Total Amount' for the bottom form is RM 630.00.

SECTION D (CONCLUSION AND REFLECTION)

4.1 Conclusion

Introducing the salary update module will significantly improve Koperasi KADA's salary deduction management. The current system's restriction, which limits flexible adjustments to salary deductions based on member needs, has become an issue for management efficiency and member satisfaction. This module will address that issue by allowing the admin to dynamically change deductions for savings and loan repayments based on member demands, hence increasing the system's flexibility and scalability. In addition to making the system more flexible to accommodate a range of financial requirements, this module will enhance member satisfaction and trust with Koperasi KADA, hence improving the member experience. At the same time, the management will become simpler and more visible, with complete records of all changes for traceability, hence eliminating human intervention and errors. To summarise, the salary update module is an effective enhancement to the existing system, allowing Koperasi KADA to better serve its members while improving overall operational efficiency, thus offering solid support for future growth.

4.2 Reflection

Our group project involves developing a web-based system for Koperasi KADA named Koperasi KADA Online System. This system aims to improve efficiency in their manual system, such as member registration, loan application and data maintenance. The core modules were the employee, member and admin. Each module serves a specific user group, addressing the operational needs of the cooperative. The employee module simplifies membership application, while the member module enables members to manage their profiles, financial status, and loan applications. The admin module allows the admin to approve memberships and loans and generate the report for decision-making. During this project, I recognized the need for an extra salary update module that would allow admin to dynamically adjust salary deductions for saving and loan repayment, as the old manual method was inflexible. This module was proposed to address specific member requirements while also improving usability. Our team was made up of five people, each of whom was in charge of specific parts of the system. I worked on the member module, which focuses on financial status and profile maintenance. Other contributed functionality to the admin module, such as membership and loan approvals and report production, as well as the employee module, which handles membership applications. While I focused on creating and testing the member module, my teammates provided feedback, fixed issues and helped with coding and documenting. Our collaboration and role distribution were critical in meeting Koperasi KADA's needs efficiently.

During this project, I had a mix of excitement and pressure. Excitement since it was a real-world product that related to our daily where I could apply my skills, but also pressured due to the system's complexity and tight deadlines. We encountered communication issues among the team at times, resulting in some confusion about task distribution and project needs. The team members' various ability levels also required us to often assist one another, which sometimes delayed our progress. Despite these challenges, I felt proud of our teamwork and the progress we made.

The project has both positive and negative aspects. On the positive side, it was fulfilling to work on a real-world project for Koperasi KADA and contribute to solving their problems. The team successfully created modules that decreased processes and enhanced flexibility, such as member registration, and loan application. However, challenges evolved, such as tight

deadlines, misunderstandings and trouble connecting modules, resulting in dissatisfaction and delays. The facts and emotions from the early stage influenced us to take a more proactive approach, such as clarifying requirements, strengthening collaborations and avoiding redundant work or misunderstandings. External variables such as limited resources schedule conflicts and the complexities of some activities influenced our choices and behaviors. At times, we had to prioritize high-priority tasks while temporarily setting aside others. These issues emphasized the importance of teamwork, particularly when dealing with technological challenges or task assignments. Feedback enabled us to constantly modify the system to fulfil the client's requirements while indicating areas for improvement, such as time management and risk assessment.

I selected this incident because it was an important moment in which we faced both success and obstacles. It involves developing a new module that is crucial to the system's functionality and has a direct impact on the client's operation. Reflecting on this situation helped me understand the value of adaption and proactive problem-solving in project development. The incident makes sense since it connects the difficulties we encountered during the project, such as communication concerns and the integration of a new module. These problems, combined with the emotions of stress and urgency, influenced our behaviors and answers, making us more aware of our work processes and the need for clear communication in future projects.

I found that, while technical ability and functionality are necessary, clear communication is critical to success. It also emphasized the importance of flexibility when dealing with unexpected challenges, and how changing our approach can lead to greater results. To avoid bottlenecks, I realize the value of having a clear project strategy, communicating constantly, and resolving difficulties early on. This project showed me the importance of adaptability and effective teamwork, particularly when incorporating new features into an existing system.

Next time, I'll prioritize proactive communication and establish clear expectations with my team and the client from the start. I'll make certain that the project timeframe is well-defined with buffer periods for unexpected complications. To effectively track progress and allocate work, I'll need better project management tools. When faced with technical issues, I will also require help from experienced mentors or coworkers, as well as further communication and conflict resolution training to effectively manage team dynamics.